Topological Spaces

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1 Motivation

The central physical object of relativistic physics is *spacetime*: a four-dimensional topological manifold that carries a smooth atlas, a torsion-free connection compatible with a Lorentzian metric, and a time orientation, and which satisfies the Einstein equations. The Einstein equations relate the matter content of the universe (one side) to gravity, encoded in the curvature of spacetime (other side).

This course clarifies each term in that definition. The first lecture covers *topological*; the second covers *manifold* and *dimension*.

Why topology? At the coarsest level, spacetime is a *set* M of points. A set alone is not enough structure to talk about **continuity of maps**. In classical physics, curves do not jump: a particle's trajectory cannot suddenly continue in a totally different place. So we need to be able to say when a curve is continuous.

Many structures would allow this (e.g. a distance measure), but we want to be minimal and not introduce undue assumptions. The weakest structure that can be placed on a set to define continuity of maps is a **topology**.

2 Topologies

Definition 1 (Topology)**.** Let M be a set. A *topology* $\mathcal{O}$ on M is a subset $\mathcal{O} \subseteq \mathcal{P}(M)$ of the power set (the set of all subsets of M) satisfying three axioms:

(i) $\emptyset \in \mathcal{O}$ and $M \in \mathcal{O}$.

(ii) If $U, V \in \mathcal{O}$, then $U \cap V \in \mathcal{O}$ (closed under finite intersections).

(iii) If $U_\alpha \in \mathcal{O}$ for all α in an *arbitrary* index set (possibly uncountable), then $\bigcup_\alpha U_\alpha \in \mathcal{O}$.

So a topology is a set of sets: a choice of certain subsets of M. The letter $\mathcal{O}$ stands for "open".

2.1 Examples on small sets

Let $M = \{1, 2, 3\}$.

Example 1. $\mathcal{O}_1 = \{\emptyset, \{1, 2, 3\}\}$ is a topology. Every intersection and union of its elements is again $\emptyset$ or M.

Example 2. $\mathcal{O}_2 = \{\emptyset, \{1\}, \{2\}, \{1, 2, 3\}\}$ is *not* a topology, since $\{1\} \cup \{2\} = \{1, 2\} \notin \mathcal{O}_2$ (axiom (iii) fails).

Remark. A topology need not be countable, even if M is. For $M = \mathbb{N}$ the power set $\mathcal{P}(\mathbb{N})$ has the cardinality of $\mathbb{R}$, and $\mathcal{O} = \mathcal{P}(\mathbb{N})$ is a topology.

2.2 Two extreme topologies

On *any* set M one can define:

- the **chaotic topology** $\mathcal{O} = \{\emptyset, M\}$ (fewest possible elements);
- the **discrete topology** $\mathcal{O} = \mathcal{P}(M)$ (maximal number of elements).

Both are generally useless in practice, but they are valuable extreme cases for testing which notions depend on what.

2.3 The standard topology on $\mathbb{R}^d$

On $\mathbb{R}^d = \mathbb{R} \times \cdots \times \mathbb{R}$ (the set of d-tuples of real numbers) there are many possible topologies. The one usually chosen (implicitly) is the standard topology. Since it has uncountably many elements, it is defined implicitly in two steps.

Step α: soft balls. For $p \in \mathbb{R}^d$ and $r \in \mathbb{R}^+$ define

$$B_r(p) = \Big\{ q \in \mathbb{R}^d \;\Big|\; \sum_{i=1}^{d} (q^i - p^i)^2 < r^2 \Big\}.$$

This is just a set of tuples; no vector space structure or norm is needed to write it down.

Step β. A set $U \subseteq \mathbb{R}^d$ is in the *standard topology* $\mathcal{O}_{\text{standard}}$ if and only if

$$\forall\, p \in U \;\; \exists\, r \in \mathbb{R}^+ : \quad B_r(p) \subseteq U.$$

Intuition in $\mathbb{R}^2$. A region whose boundary is *not* included is open: around each point one can find a small enough ball that stays inside. If some boundary points are included, then at such a point any ball, however small, sticks out of the set, so the set is not open.

Checking that this satisfies the three axioms is a short exercise.

Warning. Nothing is mathematically special about the standard topology; it is only the first one you met. It can only be defined on $\mathbb{R}^d$, not on an arbitrary set. Whether spacetime should carry it is a physical question. Whenever $\mathbb{R}$ or $\mathbb{R}^d$ appears without further comment, the standard topology is assumed.

2.4 Terminology

- A set M together with a topology $\mathcal{O}$, i.e. the pair $(M, \mathcal{O})$, is a **topological space**.
- A subset $U \subseteq M$ with $U \in \mathcal{O}$ is called an **open set**. Openness is *defined* by the choice of topology.
- A subset $A \subseteq M$ is called **closed** if its complement $M \setminus A$ lies in $\mathcal{O}$.

Warning. "Closed" is not the opposite of "open". A set can be open and closed (e.g. $\emptyset$ and M always are), open but not closed, closed but not open, or neither.

Example 3. Let $M = \{1, 2\}$ with the discrete topology $\{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. The set $\{2\}$ is open, and also closed, because $M \setminus \{2\} = \{1\}$ is in the topology. This has nothing to do with "the boundary being included or not", although those pictures work for the standard topology.

3 Continuous maps

A **map** $f : M \to N$ takes each $m \in M$ to its image $f(m) \in N$ (M is the *domain*, N the *target*). It need not be injective (two points may share an image) or surjective (some points of N may not be hit). A map is defined between sets with no further structure.

Whether f is continuous depends, by definition, on which topologies are chosen on the domain and the target.

Definition 2 (Pre-image)**.** For $f : M \to N$ and a subset $V \subseteq N$, the *pre-image* is

$$\mathrm{preim}_f(V) = f^{-1}(V) = \{m \in M \mid f(m) \in V\} \subseteq M.$$

This is defined for any subset V and does not require f to be invertible, nor any topology.

Definition 3 (Continuity)**.** Let $(M, \mathcal{O}_M)$ and $(N, \mathcal{O}_N)$ be topological spaces. A map $f : M \to N$ is *continuous with respect to $\mathcal{O}_M$ and $\mathcal{O}_N$* if for every $V \in \mathcal{O}_N$,

$$\mathrm{preim}_f(V) \in \mathcal{O}_M.$$

Mnemonic: *pre-images of open sets are open sets* (open in the target, pulling back to open in the domain). No ε's and δ's are needed.

Remark. The pre-image of the empty set is always the empty set, because every element of the domain is mapped to something. It is important that $\emptyset \in \mathcal{O}_M$; otherwise only surjective maps could be continuous.

3.1 Example: continuity depends on the topologies

Let $M = N = \{1, 2\}$ and let $f : M \to N$ swap the points: $f(1) = 2$, $f(2) = 1$. Choose

$$\mathcal{O}_M = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}, \qquad \mathcal{O}_N = \{\emptyset, \{1, 2\}\}.$$

(a) $f : (M, \mathcal{O}_M) \to (N, \mathcal{O}_N)$ **is continuous.** The only open sets in N are $\emptyset$ and $\{1, 2\}$, with pre-images $\emptyset$ and M, both in $\mathcal{O}_M$.

(b) $g = f^{-1} : (N, \mathcal{O}_N) \to (M, \mathcal{O}_M)$ **is not continuous.** Here g also swaps 1 and 2, but now $\{1\} \in \mathcal{O}_M$ is open in the target with pre-image $g^{-1}(\{1\}) = \{2\}$, which is not in $\mathcal{O}_N$.

So a continuous bijection can have a discontinuous inverse. Even the identity map between the same set with two different topologies can be continuous in one direction but not in the other. It is dangerous not to state the topologies explicitly.

3.2 Homeomorphisms

A bijective map that is continuous in both directions is called a **homeomorphism**. Then not only pre-images but also images of open sets are open, so homeomorphisms preserve the topological structure: they are the structure-preserving maps (isomorphisms) of topological spaces.

4 Composition of continuous maps

For $f : M \to N$ and $g : N \to P$, the composition $g \circ f : M \to P$ is defined by $(g \circ f)(m) = g(f(m))$.

Theorem 1. *If $f : M \to N$ and $g : N \to P$ are continuous (with respect to topologies $\mathcal{O}_M, \mathcal{O}_N, \mathcal{O}_P$), then $g \circ f$ is continuous.*

Proof. Let $V \in \mathcal{O}_P$. Then

$$\begin{aligned}\mathrm{preim}_{g\circ f}(V) &= \{m \in M \mid g(f(m)) \in V\} \\ &= \{m \in M \mid f(m) \in \mathrm{preim}_g(V)\} \\ &= \mathrm{preim}_f\big(\mathrm{preim}_g(V)\big).\end{aligned}$$

Since g is continuous and V is open, $\mathrm{preim}_g(V) \in \mathcal{O}_N$. Since f is continuous, the pre-image of this open set is in $\mathcal{O}_M$. Hence $\mathrm{preim}_{g\circ f}(V) \in \mathcal{O}_M$. □

This uses only the definition of continuity, with no ε–δ arguments.

Curves. A curve is a map from a parameter space (e.g. $\mathbb{R}$, with the standard topology) into the space in which it runs (which also needs a topology). "No jumps" then translates to "the curve is continuous".

5 Inheriting a topology: the subset topology

There are many useful ways to construct a new topology from given topological spaces. The one most important for spacetime physics is the following.

Definition 4 (Subset topology)**.** Let $(M, \mathcal{O}_M)$ be a topological space and $S \subseteq M$ any subset. The *subset topology* on S (inherited from M) is

$$\mathcal{O}_M|_S = \{U \cap S \mid U \in \mathcal{O}_M\} \subseteq \mathcal{P}(S).$$

Theorem 2. *$\mathcal{O}_M|_S$ is a topology on S.*

Proof. (i) $\emptyset = \emptyset \cap S$ with $\emptyset \in \mathcal{O}_M$, and $S = M \cap S$ with $M \in \mathcal{O}_M$; so both lie in $\mathcal{O}_M|_S$.

(ii) Let $A, B \in \mathcal{O}_M|_S$. Then $A = \tilde{A} \cap S$ and $B = \tilde{B} \cap S$ for some $\tilde{A}, \tilde{B} \in \mathcal{O}_M$. Then

$$A \cap B = (\tilde{A} \cap S) \cap (\tilde{B} \cap S) = (\tilde{A} \cap \tilde{B}) \cap S,$$

and $\tilde{A} \cap \tilde{B} \in \mathcal{O}_M$, so $A \cap B \in \mathcal{O}_M|_S$.

(iii) For unions the argument is the same: $\bigcup_\alpha (\tilde{A}_\alpha \cap S) = \big(\bigcup_\alpha \tilde{A}_\alpha\big) \cap S$. □

Why it matters. Suppose $f : M \to N$ is continuous (say it is easy to see this for the given topologies), and we only care about the restriction $f|_S : S \to N$ to a subset $S \subseteq M$. For example, $f : \mathbb{R}^2 \to \mathbb{R}$ could be a temperature distribution on a hot plate, and S a wire on the plate along which a mouse can move; only the temperature along the wire matters.

If S is given an arbitrary topology, continuity of $f|_S$ cannot be concluded. But if S carries the subset topology inherited from M, then $f|_S$ is guaranteed to be continuous. Indeed, for $V \in \mathcal{O}_N$,

$$\mathrm{preim}_{f|_S}(V) = \mathrm{preim}_f(V) \cap S,$$

which lies in $\mathcal{O}_M|_S$ because $\mathrm{preim}_f(V) \in \mathcal{O}_M$.

Summary

- A **topology** on a set is the weakest structure allowing a definition of continuity; it is a choice, not automatic.
- Open sets are *defined* as the members of the chosen topology; closed sets are complements of open sets.
- f is **continuous** iff pre-images of open sets are open.
- Compositions of continuous maps are continuous.
- Subsets inherit a topology under which restrictions of continuous maps stay continuous.